COMPARISON OF PERFORMANCE-BASED INDICATORS: INDIA & CHINA

DEPARTMENT OF ECONOMICS

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Abstract

India and China, the world's most populous nations, have pursued distinct paths of development shaped by diverse policies and reforms. This study examines key aspects such as population dynamics, health, and economic progress, highlighting shared challenges and unique strategies that have influenced their growth trajectories and future potential.

Keywords: Population dynamics, demographic shifts, Economic development, health outcomes, gender disparities, dependency ratio

INTRODUCTION

India and China, together accounting for more than one-third of the world's population, have followed interesting yet divergent paths in their journey towards development. From massive populations to fast-growing economies and health improvements to managing demographic shifts, this paper looks at the two global powerhouses. It looks at key areas like population, health, and the economy to show how different policies, cultural changes and global events have shaped their futures.

Using historical data, charts, and future projections, our analysis highlights both the similarities and differences in how India and China have evolved. It explains the role of economic reforms, social changes, and government decisions in impacting growth and development. This analysis also tries to look into the future challenges and opportunities that both countries may face, owing to their different demographic and economic situations.

1. POPULATION

1.1. Comparative Trends in Population Growth Rates

1.1.1. Analysing Past Trends:

The graph shows the population trends[1] of India and China from 1960 to 2016. While both countries experienced steady growth, China's line appears flatter, indicating a slower growth rate compared to India's upward trend. This difference reflects the impact of varying population policies and demographic dynamics over time.

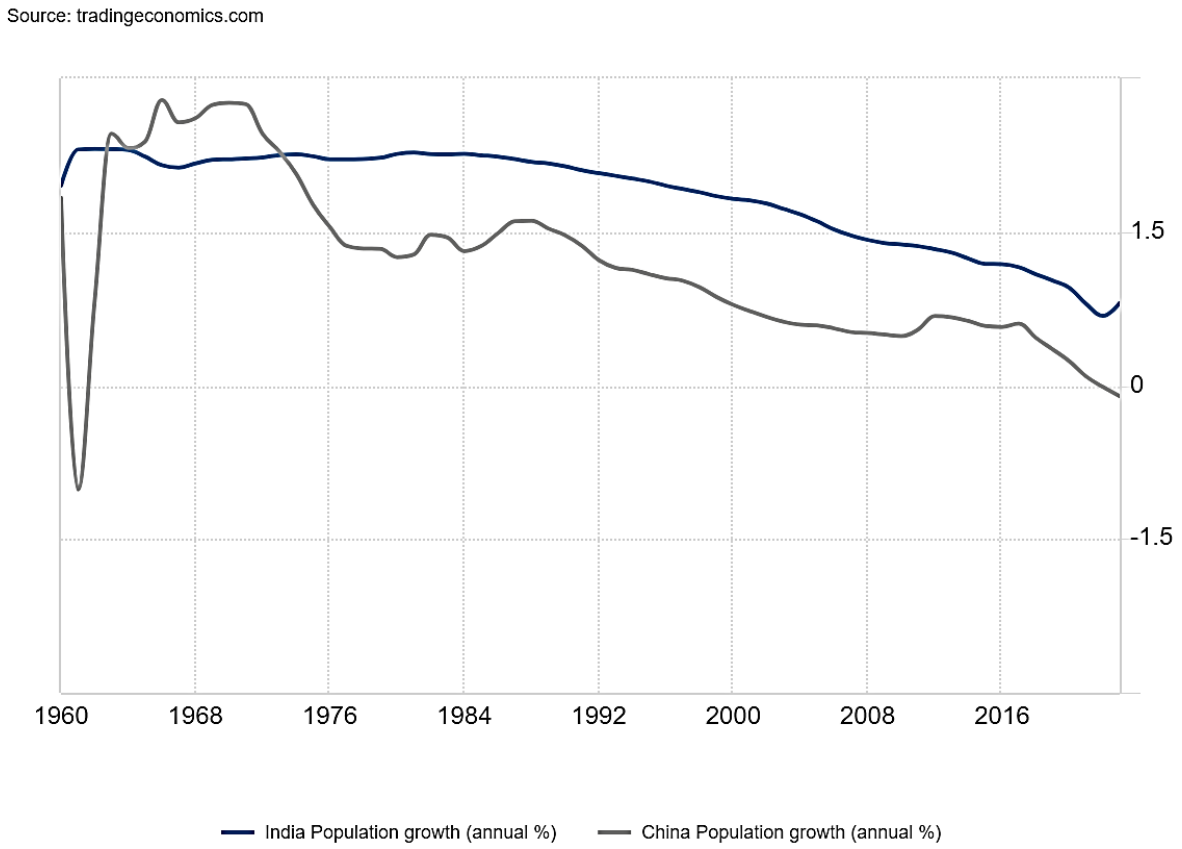


Fig 1(a): Trends in Population Growth Rates
(Source: World Bank)

Both countries experienced a slowdown in population growth compared to earlier decades due to government efforts to control population expansion. While growth rates decreased, both nations

[1] Population growth rate is computed by taking the current population size and subtracting the previous population size and dividing that amount by the previous size. Subsequently, the annual growth rate can be calculated by dividing that by the number of years between the current and previous observations.

followed a trend towards gradual stabilization. Despite different trajectories, the slowing pace reflects shared challenges of managing large populations.

China's population growth rate flattened earlier, largely due to the "One-Child Policy" implemented in 1979. This policy significantly curbed birth rates, shaping a demographic profile characterized by lower growth and a rising elderly population. In contrast, India's growth remained more gradual due to higher fertility rates and a demographic momentum carried forward from previous decades. Improved healthcare and lower infant mortality also contributed to India's continued population increase.

China's strict population control measures and rapid economic development contributed to its lower growth rate. The relaxation of these measures around 2015 was intended to address an aging population and potential labour shortages. On the other hand, India's higher population growth resulted from "population momentum" and a relatively younger demographic structure, sustaining higher birth rates.

1.1.2. Estimated Future Trends:

The differences between India and China's population trends are expected to diverge further. As of 2024, India's population exceeded China's, growing at 0.90% annually, while China's population began declining in 2021 and could fall below 1 billion by 2070. India is projected to maintain growth until around 2062, potentially benefiting from a demographic dividend, while China's demographic challenges include an aging population and a shrinking workforce. These trends may continue to widen the gap in the future.

While both countries have experienced a slowdown in population growth, India's demographic momentum is likely to keep its growth rate higher than China's, resulting in significant shifts in the global demographic landscape.

1.2. Comparative Trends in Dependency Ratios

1.2.1. Analysing Past Trends:

The dependency ratio[2] or India has fallen from 66% in 2001 to below 50% in recent years and is projected to go further down. One may consider a population to be in the "dividend phase" if the

[2] Age dependency ratio is the ratio of dependents, people younger than 15 or older than 64, to the working-age population--those ages 15-64. Graphs are shown as the proportion of dependents per 100 working-age population.

share of the working age population is 67% or higher. Thus, India is already in the dividend phase. However, the "potential dividend" is yet to be harvested.

It is to be noted that the positive effect of a rising support ratio on economic growth can be magnified by ensuring that women gain greater access to paid work and their time in poorly remunerated activities is minimised. Also, Indian educational institutions have failed to take advantage of the opportunity to build human capital in our students. Unless this can be redressed, it would be difficult to take full advantage of the demographic dividend (Sonalde Desai, 2014).

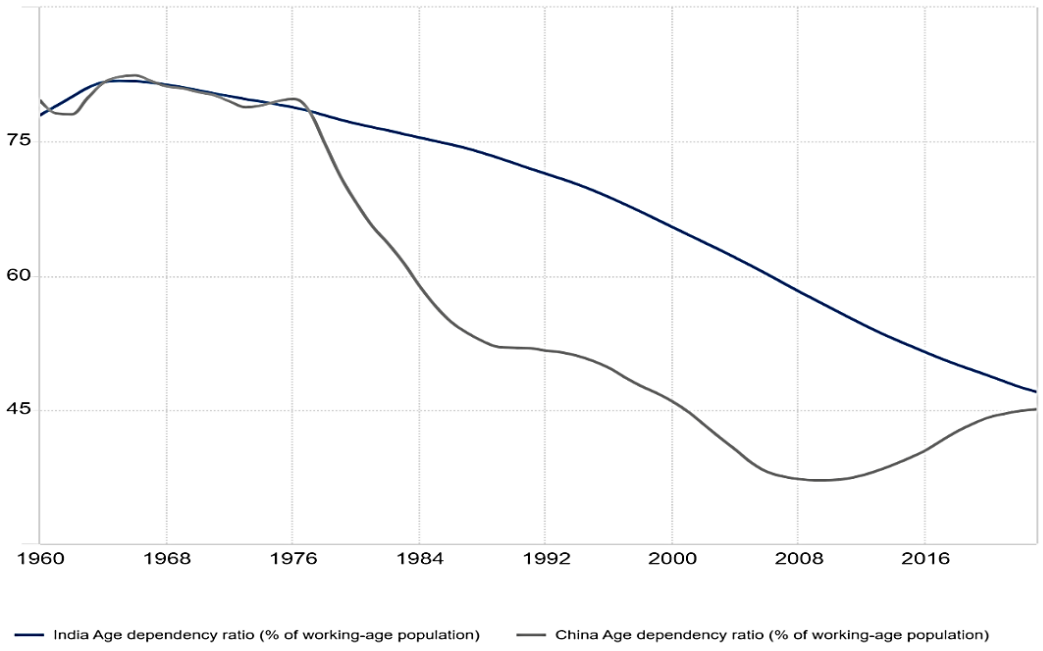


Fig 1(b): Trends in Dependency Ratios
(Source: World Bank)

In the past few decades, China's economy has profited from a relatively low dependency ratio, accompanied with a growing working age population. These two demographic causes accounted for China's large and cheap labour force. The fall in dependency ratio since the 1970s has been mainly due to lower birth rates and a resulting decrease of child dependency. This led the dependency ratio to reach a historic low (below 40%) between 2005 and 2017.

A turning point was reached around the year 2010, when the effects of declining child dependency were neutralized by growing old-age dependency.

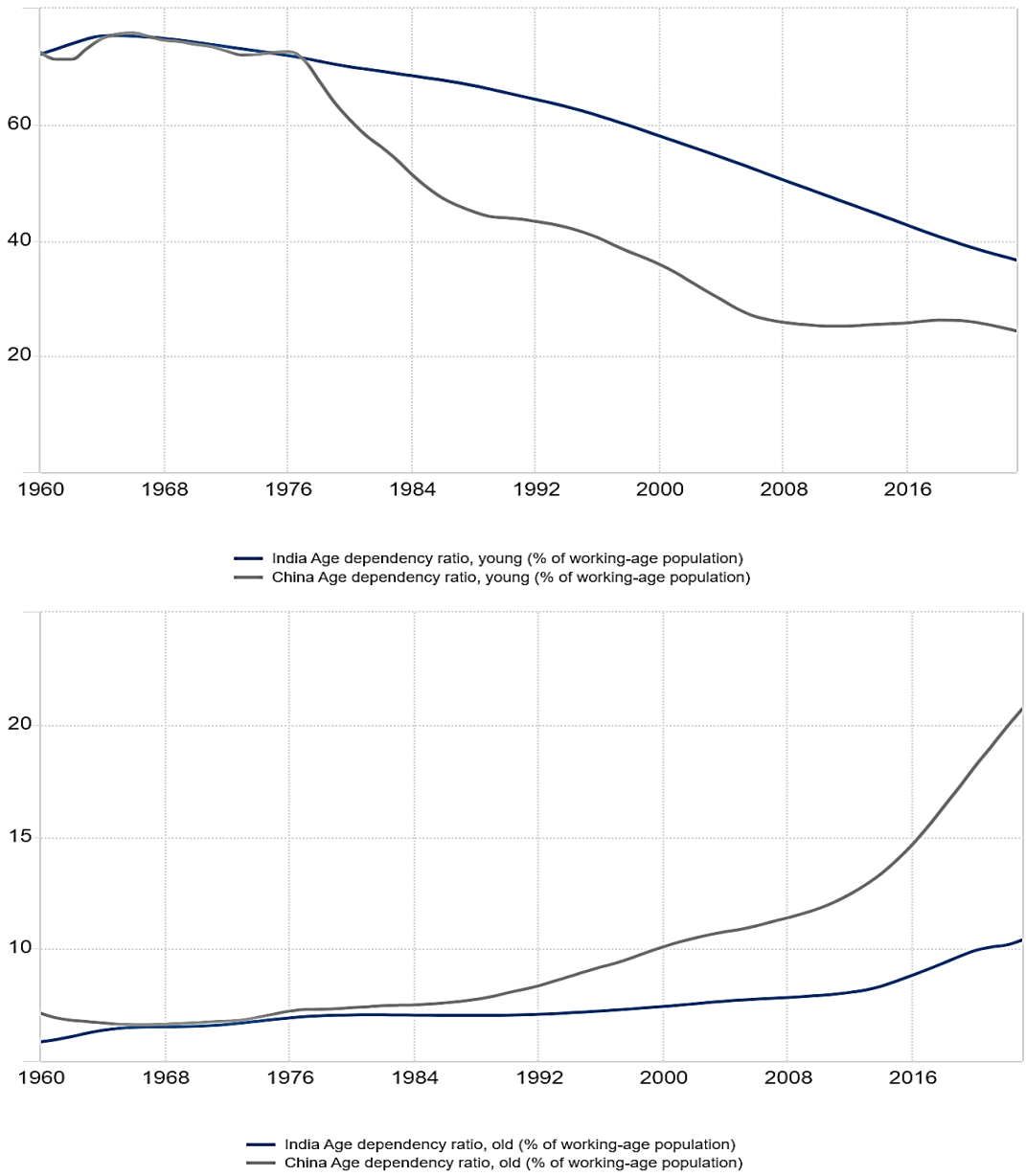


Fig 1(c): Trends in Dependency Ratios for
Old and Young Population
(Source: World Bank)

1.2.2. Estimated Future Trends:

China's population is aging very rapidly. The Chinese state media reports that China is already approaching a "moderately aging" scenario with 20% of its population being aged 60 years and

older. By 2035, that percentage is projected to rise to 30%. According to Pew Research Center, China also appears poised to roughly double its “dependency ratio” by 2100.

On the other hand, India’s dependency ratio is projected to be at its lowest by 2030. India’s young dependency ratio is expected to overtake the old dependency ratio. India would remain the largest provider of human resources in the world. However, soon after that, India may have to face the challenge of ageing population as well, as the current middle-aged working population would become the old-age population by then.

2. SOCIETY

2.1. Comparative Trends in Sex Ratios:

2.1.1. Analysing Past trends:

The sex ratio in India and China has been a significant demographic issue, shaped by cultural, social, and economic factors. As of the latest census (2021), India has a sex ratio of about 1,020 males for every 1,000 females whereas that of China has been notably skewed, with estimates showing about 1,050 males for every 1,000 females, especially among younger age groups.

A combination of cultural preferences, government decree and modern medical technology in the world’s two largest countries has created a gender imbalance on a continental scale. Men outnumber women by 70 million in China and India. Although China is generally more pronounced, particularly in the younger population.

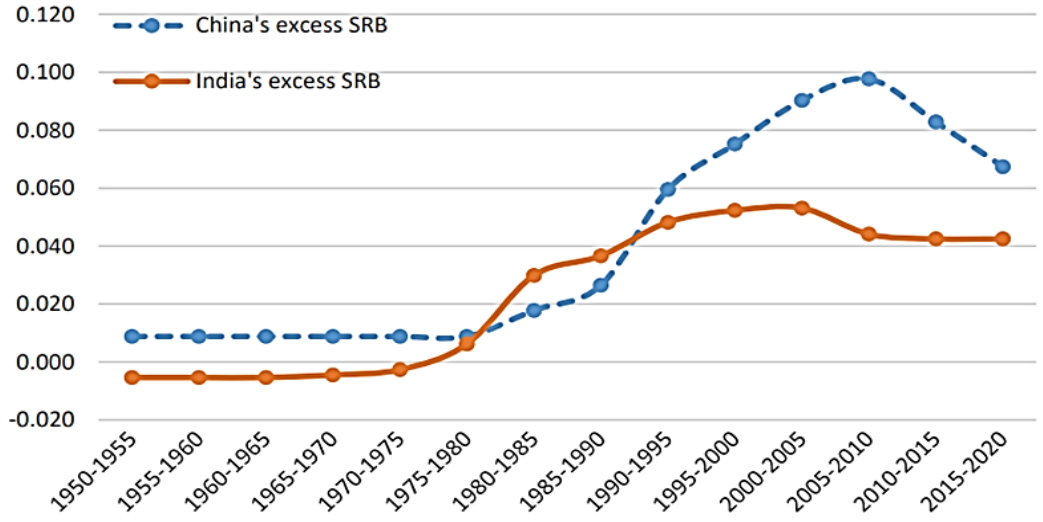


Fig 2(a): Comparative Trends in Sex Ratio at Birth
(Source:World Bank)

The Indian government has launched initiatives like the "Beti Bachao, Beti Padhao" (Save the Girl Child, Educate the Girl Child) to promote gender equality and address female infanticide and gender discrimination. However, China has shifted policies to encourage more balanced family structures. China's One-Child Policy inadvertently intensified gender selection practices. Families, restricted to one child, often opted for a boy, leading to an increase in sex-selective abortions. Although the policy has been relaxed, the legacy of skewed sex ratios persists.

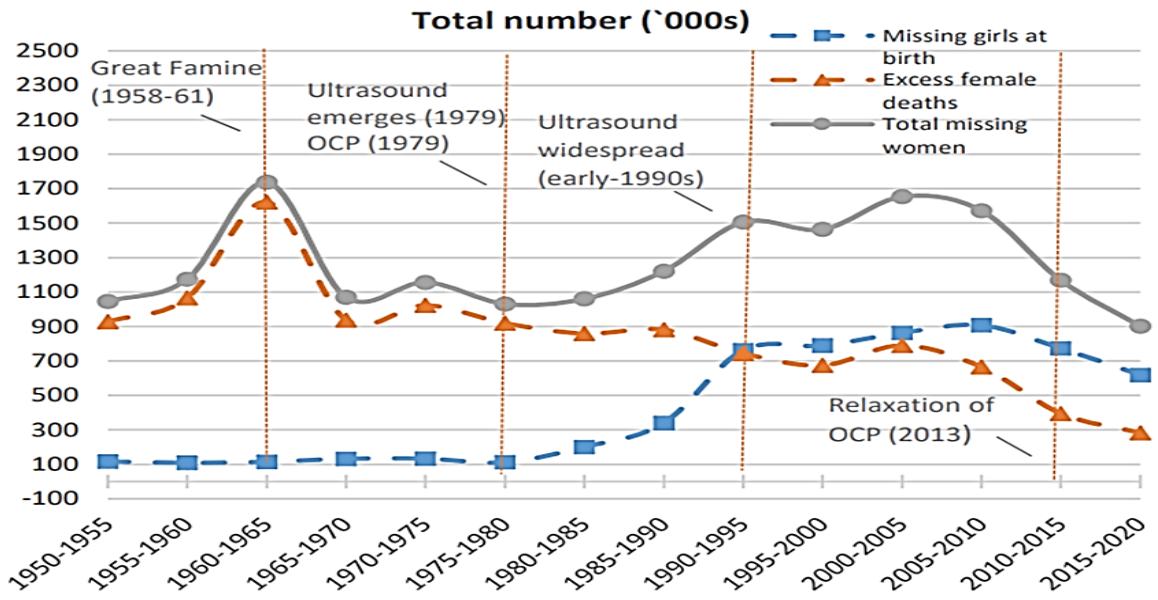


Fig 2(b): Missing Women- China
(Source:World Bank)

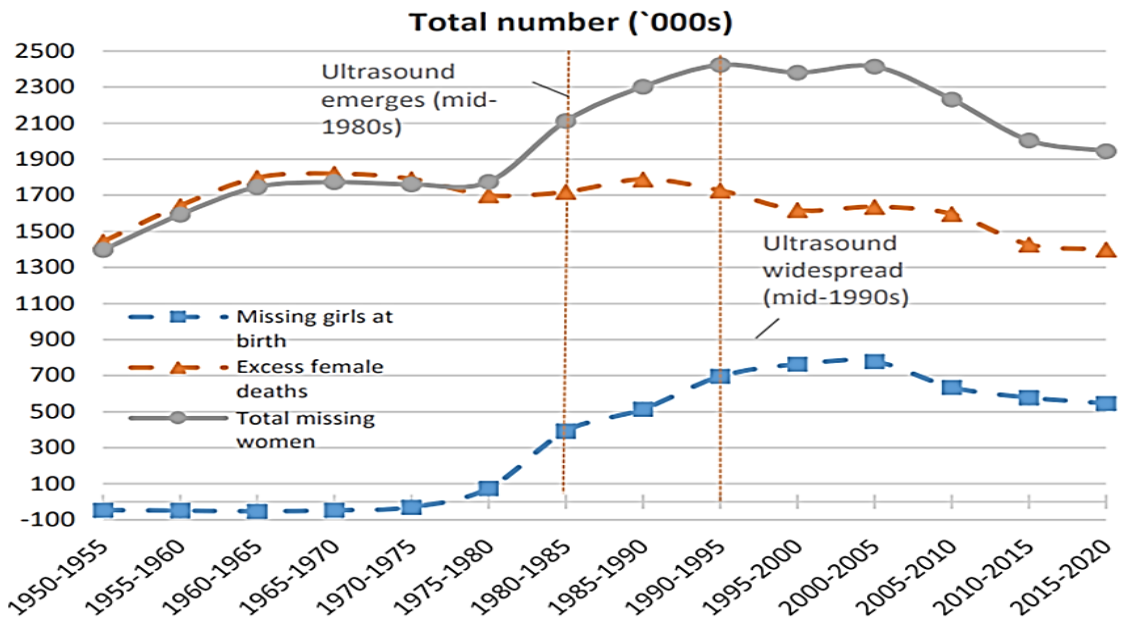


Fig 2(c): Missing Women- India
(Source:World Bank)

2.1.2. Estimated Future Trends:

However, while both India and China experience significant sex ratio imbalances driven by cultural, economic, and policy factors, the magnitude and implications differ. Both countries are making efforts to address these challenges, but persistent societal norms and historical legacies continue to influence gender preferences. Addressing these issues holistically requires continued commitment to gender equality, education, and policy enforcement.

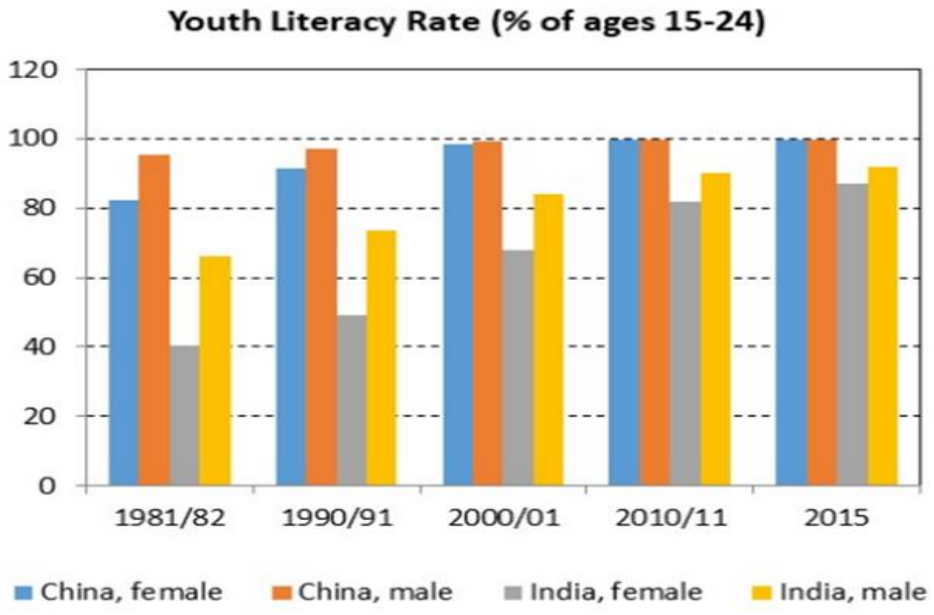
2.2. Comparative Trends in Literacy Ratios:

2.2.1. Analysing Past trends:

The literacy ratios in India and China reflect the historical, social, and economic contexts of both countries. China is found to be far ahead of India in terms of Education. As of recent estimates, China has a literacy rate of approximately 97-99%, with significant efforts made in recent decades to ensure universal education. While India’s literacy rate stands around 77-80%, with disparities between urban and rural areas, as well as among different states and genders.

In terms of Historical context, China has made substantial strides in literacy since the mid-20th century, particularly after the establishment of the People's Republic of China in 1949, which prioritized education as a means of social and economic development. Campaigns like the "Cultural Revolution" aimed at promoting basic literacy contributed to rapid gains. While India's literacy efforts gained momentum post-independence in 1947, but the progress has been uneven due to diverse linguistic, cultural, and socioeconomic factors. Initial investments in education faced challenges related to poverty, gender disparities, and access in rural areas.

In the context of Gender disparities, China’s literacy gap between the two genders have narrowed significantly, with female literacy rates approaching those of males. However, Gender disparities remain more pronounced in India, with female literacy rates significantly lower than male rates. Factors such as cultural norms, early marriage, and poverty affect girls' education more acutely.



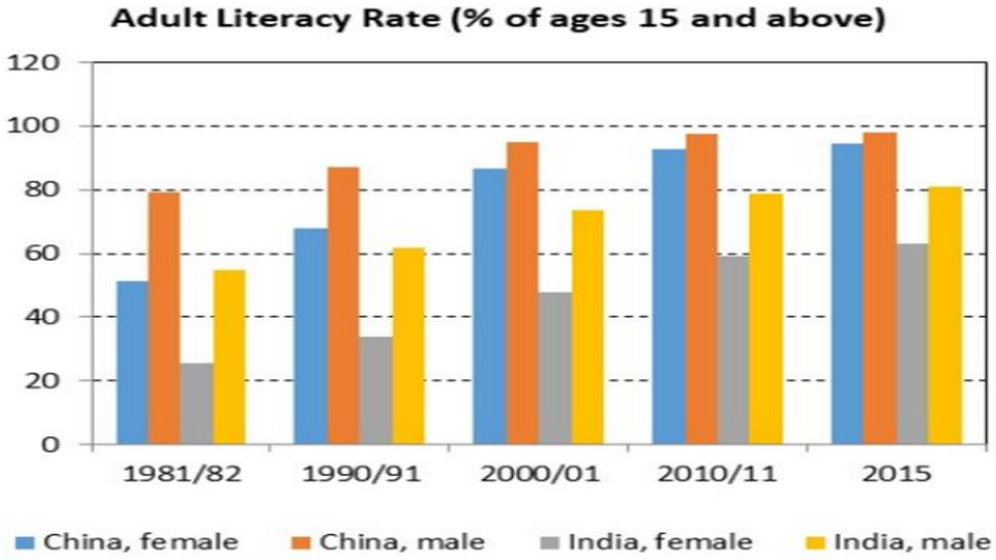


Fig 2(d): Comparative Trends in Literacy Rates
(Source: World Bank)

In India there is a stark contrast between urban and rural literacy rates, with urban areas achieving higher literacy levels. Rural areas often experience inadequate infrastructure and resources, leading to lower educational attainment.

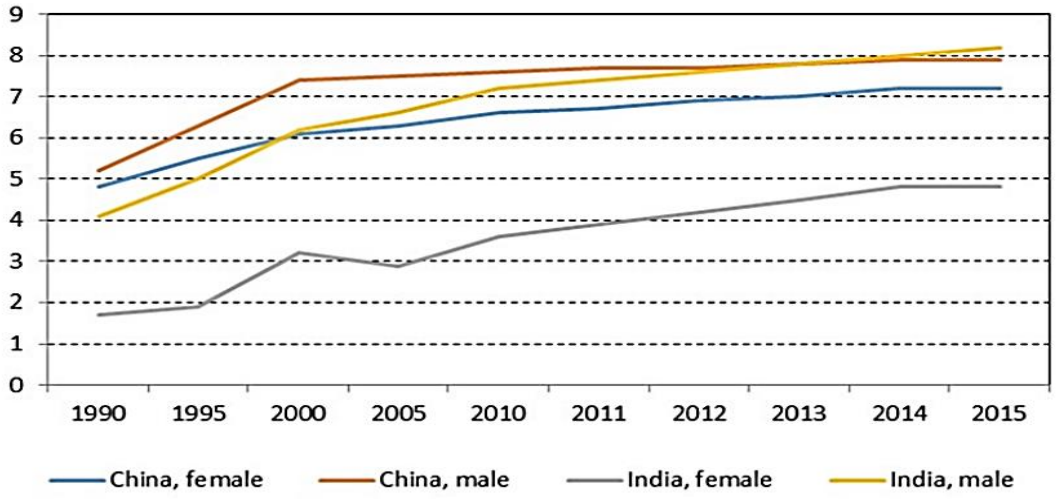


Fig 2(e): Comparative Trends in Mean Years of Schooling
(Source: HDR 2017)

2.2.2. Estimated Future Trends:

Thus, both India and China have made significant strides in literacy, but China is poised to maintain its high literacy levels, with a focus shifting toward enhancing educational quality and developing vocational skills. In contrast, India's literacy rates are likely to improve through ongoing government initiatives and growing social awareness, although challenges such as regional disparities, gender inequalities, and educational quality will need to be addressed to ensure sustainable progress.

3. HEALTH

3.1. Comparative Trends in Infant Mortality Rates:

3.1.1. Analysing Past Trends:

The comparison of Infant Mortality Rates[3] (IMR) between India and China from 1969 to 2022 reveals contrasting trajectories in health and developmental progress. In the late 1960s, both countries had relatively high IMRs due to underdeveloped healthcare systems, poor sanitation, malnutrition, and widespread poverty. In 1969, India's IMR was approximately 140 deaths per 1,000 live births, while China's IMR was slightly lower, at around 89 deaths per 1,000 live births. However, over the decades, the two countries pursued different development strategies, leading to significant divergences in health outcomes.

[3] Infant mortality rate is the number of deaths of children under one year of age. This indicator is measured as the number of deaths per 1,000 live births.

China, after the implementation of aggressive public health campaigns, improved rural healthcare access through the "barefoot doctors" program, and later through economic reforms and rapid industrialization, achieved a drastic reduction in its IMR. By the late 1990s, China's IMR had dropped to around 30 per 1,000, and by 2022, it had further declined to about 5-7 per 1,000 live births. China's sharp decline can be attributed to strong investments in healthcare infrastructure, vaccination programs, improved maternal and child healthcare services, and efforts to eradicate extreme poverty.

India, on the other hand, faced a slower decline due to factors like a larger population, slower economic growth in the earlier decades and greater regional disparities in healthcare access. However, India also made significant strides, especially post-2000, with programs such as the National Rural Health Mission (NRHM), improving access to maternal healthcare, and initiatives like Janani Suraksha Yojana aimed at reducing childbirth-related deaths. By 2022, India's IMR had fallen to around 27-30 deaths per 1,000 live births, though still considerably higher than China's.

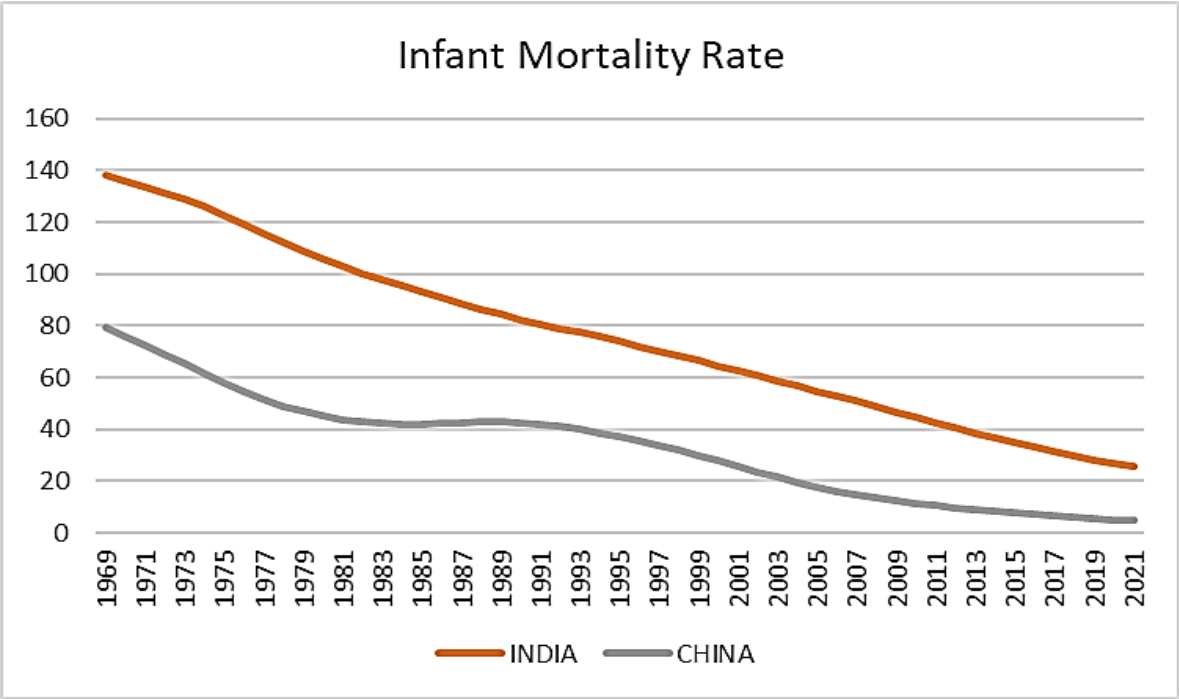


Fig 3(a): Trends in Infant Mortality Rates
(Source:World Bank Open Data)

3.1.2. Summarising Trends:

In conclusion, while both countries saw substantial reductions in infant mortality, China's progress was more rapid and consistent, largely due to earlier and more comprehensive health and economic reforms. India, though improving significantly, continues to face challenges such as regional inequality, healthcare accessibility, and high malnutrition rates that have slowed the pace of decline compared to China.

3.2. Comparative Trends in Life Expectancies at Birth:

3.2.1. Analysing Past Trends:

The comparison of life expectancy[4] between India and China from 1960 to 2022 highlights significant differences in the health and socio-economic progress of the two countries over the last six decades. In 1960, both nations had relatively low life expectancy rates due to poverty, underdeveloped healthcare systems and high rates of infectious diseases. At that time, India's life expectancy was around 45 years, while China's life expectancy was slightly lower at about 33 years. Over the following decades, both countries experienced improvements, but at different rates, driven by distinct policies and developmental paths.

China's life expectancy began to rise sharply following the Maoist era healthcare reforms and the introduction of basic medical services through programs like the "barefoot doctors", which provided healthcare to rural populations. Following the economic liberalization under Deng Xiaoping in the late 1970s, China's rapid industrialization and investments in health, sanitation, and education accelerated life expectancy gains. By 1990, China's life expectancy had increased to around 68 years. Over the next few decades, China's continued emphasis on healthcare access, combined with rising incomes, improved living standards and reduced poverty, pushed life expectancy even higher. By 2022, China's life expectancy stood at around 78-80 years, on par with many developed countries.

India's progress, while significant, was more gradual. From 1960 to 1980, India made strides in reducing infant mortality, eradicating diseases like smallpox, and improving healthcare access, but the gains in life expectancy were slower compared to China. In 1990, India's life expectancy was about 58 years, roughly 10 years behind China. The disparity can be attributed to factors such as India's larger and more diverse population, slower economic growth and uneven healthcare access,

[4] Life expectancy at birth is the average number of years a newborn is expected to live, assuming that age-specific mortality rates remain constant.

especially in rural and impoverished regions. However, from the 1990s onward, India made significant improvements through various health initiatives, including National Health Mission programs, increased vaccination coverage, better maternal and child healthcare and economic reforms that boosted living standards. By 2022, India's life expectancy had risen to around 68-70 years, a notable improvement but still about 8-10 years behind China.

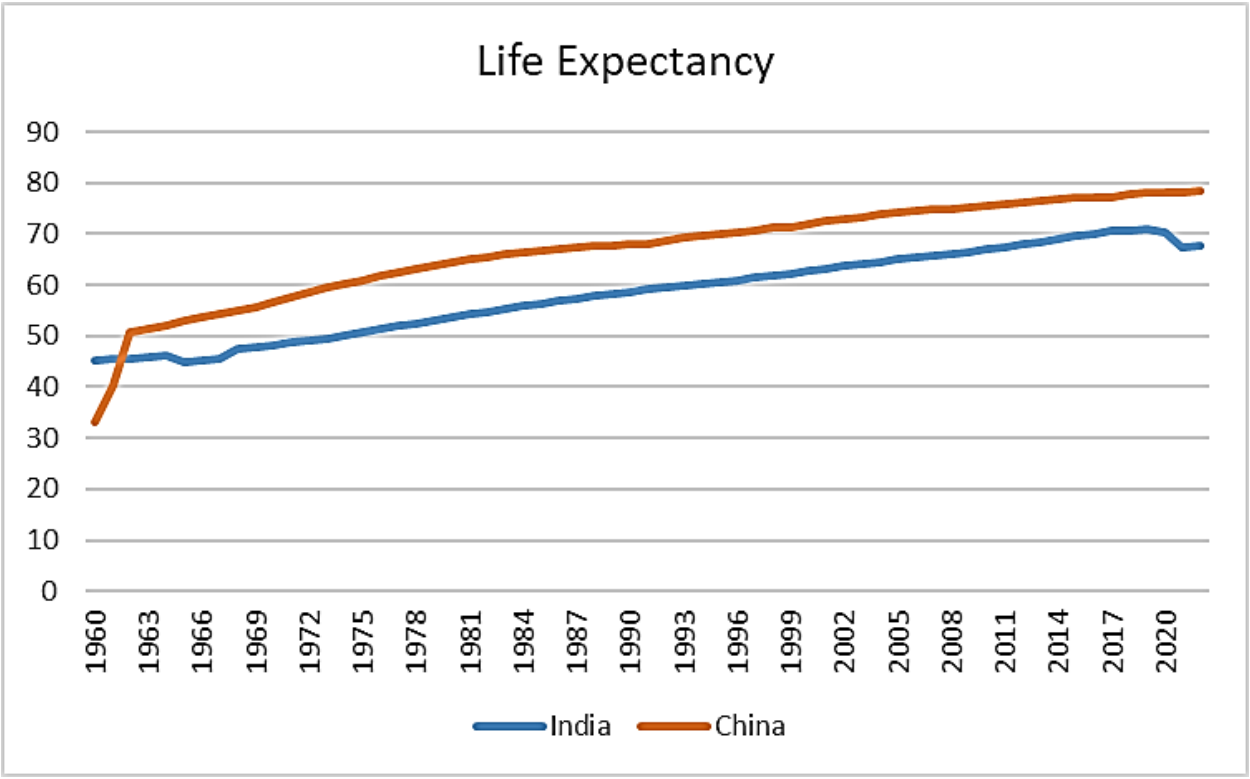


Fig 3(b): Trends in Life Expectancy
(Source:World Bank Open Data)

3.2.2. Summarising Trends:

In summary, while both India and China have seen remarkable improvements in life expectancy since 1960, China's life expectancy has consistently been higher and has grown more rapidly, thanks to earlier and more widespread healthcare reforms, rapid economic development, and a more equal distribution of healthcare services. India, despite making substantial gains, continues to face challenges related to healthcare access, regional disparities and public health infrastructure, which have contributed to its slower increase in life expectancy compared to China.

4. ECONOMY

4.1. Comparative Trends in GDP Growth Rates:

4.1.1. Analysing Past Trends:

In the early stages following the foundation of the People's Republic of China (1950), the national economy of China and India were at similar levels. In 1950, the Chinese per capita gross domestic product (GDP) was only 43.5% of India's. In the late 1970s, with the opening up of reform policies, the Chinese economy made rapid progress and notable achievements, regarded as the "Chinese miracle" (lin and Zhou 2011). Both nations have seen rapid economic expansion, with different growth patterns and phases of development. While China captured the world's attention with its double-digit growth for decades, India has steadily built its own growth momentum.

The graph below shows that India started with a slower growth rate[5] but has gradually increased its pace over the decades. While it remained behind China for most of the period, India has steadily improved its economic performance, particularly since 2000. India's upward trend in recent years signifies its potential as a growing economic powerhouse, whereas China appears to be entering a phase of slower growth.

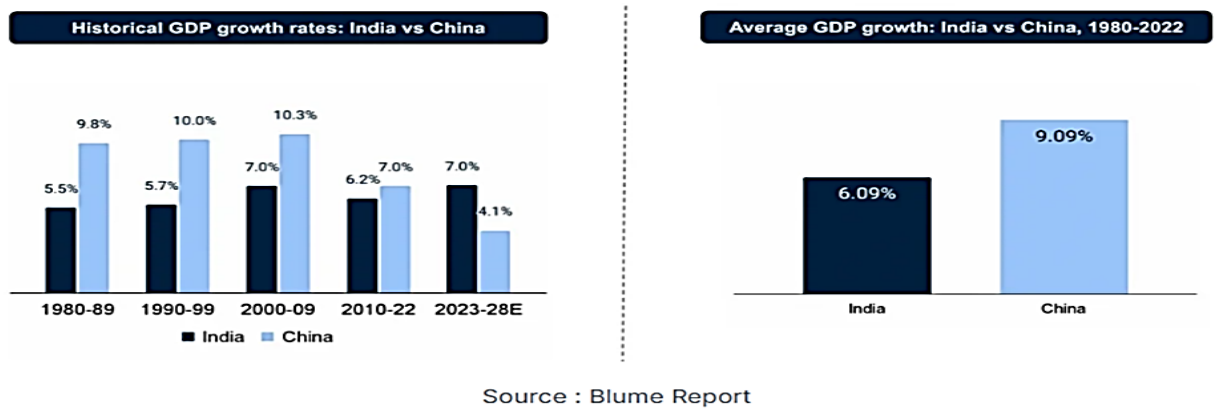


Fig 4(a): Trends in GDP Growth Rates

Both countries experienced sustained growth due to economic reforms—China's in 1978 and India's in 1991—and urbanization. Globalization and foreign investment further accelerated their rise. They also moved from primarily agrarian economies to industrial and service-driven growth.

However, from 1980 to 2010, China had comfortably outpaced India in terms of GDP growth, averaging around 10% every year and while India managed to hover between 5.5% to 7%. The strong state-planning with an export-led strategy had contributed much to China's initial

[5] Growth Rate = (GDP_Year2/ GDP_Year1) - 1

advantage, while the Indian economy was held back due to bureaucratic apathy and lack of infrastructure. As the software industry began to take off from 1986, there were quick exports from India as well. Low fertility was a potential challenge to GDP growth in the future in China, coupled with the associated ageing of the population, thus proposing that growth in the GDP will suffer as an outcome. India, by contrast, entered the millennium with much more youthful population than China.

4.1.2. Estimated Future Trends:

Growth rates of GDP can converge or diverge because of a complex interplay of various factors. China's old maturing economy and challenges in demographics might make the growth rates converge, but better infrastructure and technological investments are another case that may keep it ahead. A younger population and an increasing service sector can offer prospects of catch-up for India, but persistent problems in infrastructure and skill development might hold back the speed of convergence. The deep-rooted, anti-meritocratic caste system, extended bureaucracy also made economists believe it would never become a global powerhouse like China.

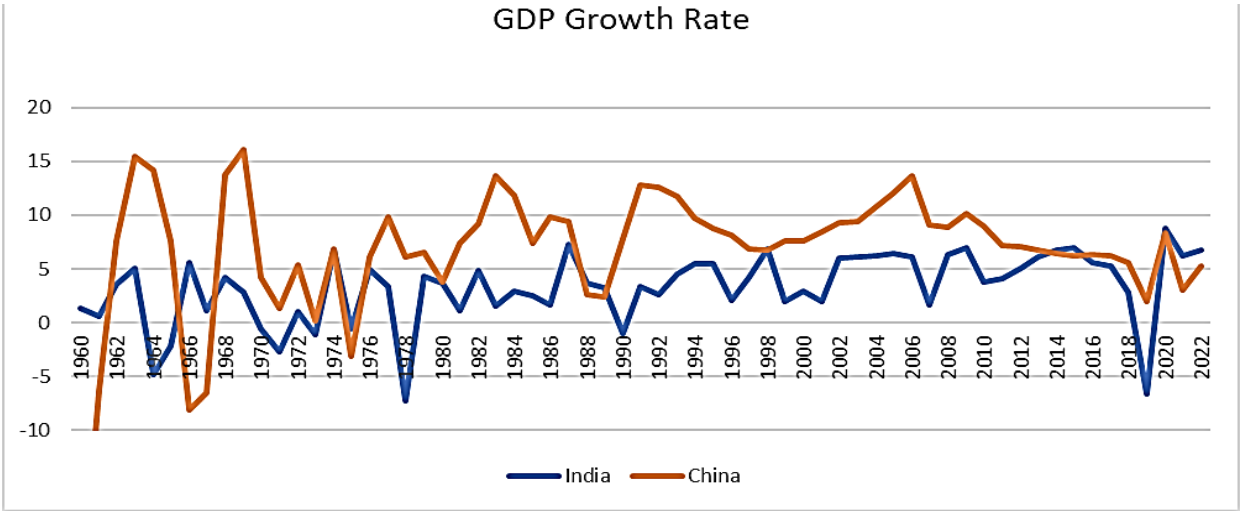


Fig 4(b): Historical trends in GDP Growth Rates
(Source: World Bank)

4.2. Comparative Trends in Unemployment Rate

4.2.1. Analysing Past Trends:

Comparing the past trends in the unemployment rate[6] for India and China, we see that neither of the countries show a clear trend. However, China's unemployment rate has remained consistently below that of India.

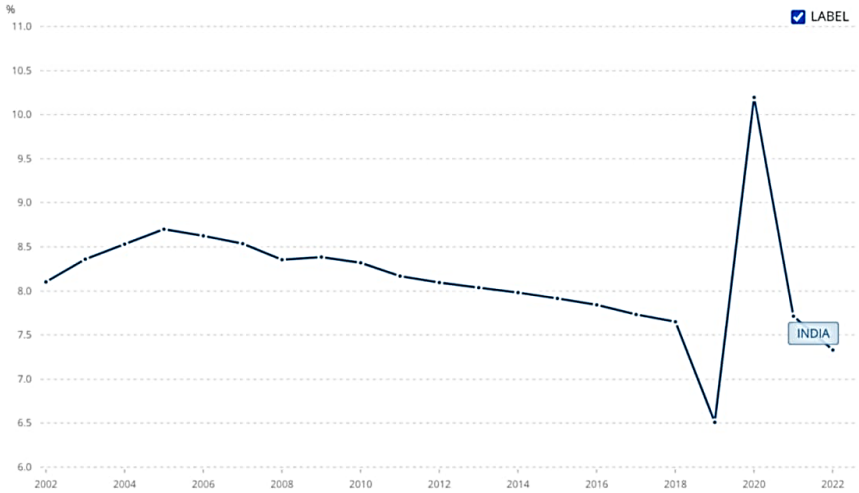


Fig 4(c): Trends in Unemployment Rates
(Source:Peng,Y. (2023))

[6] The unemployment rate is the percentage of the labor force that is looking for a job. Unemployment rate = number of unemployed persons / labor force.

China faced temporary unemployment in the beginning of 2000s because of its reform of state-owned enterprises (SOEs). Though SOEs improved efficiency and reduced overcapacity it led to layoffs resulting in sharp unemployment. India also faced persistent unemployment in 2000 because of the increasing job demand by the working population. Both China and India faced severe unemployment in 2008 because of the Global Financial crisis which affected the export sector of both the countries. However, China stabilized its economy by 2010 by investing in various projects but India continued to face unemployment because of failed education system which could not meet the industry demands. We must also note how trade war of 2018 affected unemployment in China. Both the countries again observed high unemployment during the covid 19 pandemic. Many small businesses were closed down which mainly affected the workers of the informal sector along with the migrant workers. Post pandemic the recovery was uneven and thus both the economies slowed down leading to job insecurity across the economy during 2021 to 2022.

By mid-2024, China's unemployment rate rose to 5.2%, reflecting broader economic challenges like reduced consumer demand and sluggish industrial growth. Urban areas saw a slightly higher rate of 5.3%. In contrast, India faces a more severe situation, with unemployment at 9.2% in June 2024, driven by a sharp rise in rural unemployment (9.3%) and regional economic disparities.

4.2.2. Estimated Future Trends:

China is expected to develop a well-functioning labour market and balanced economic condition thus, its unemployment is also expected to remain stable in the coming years. In contrast, the current policies of India are not enough to sustain stable job growth and there may be structural issues in labour market leading to a fluctuating unemployment rate.

Conclusion

India and China, two of the world's most populous nations, have embarked on divergent paths of development shaped by distinct policies, socio-economic contexts, and demographic trends. While China experienced earlier stabilization of population growth, economic reforms, and significant health improvements, India has witnessed sustained demographic momentum, a younger population, and a gradual rise in economic performance. Both countries have leveraged reforms to drive growth but face unique challenges, including aging populations in China and unemployment and educational disparities in India. This comparative study underscores their shared and differing trajectories, offering insights into how demographic shifts, policy interventions, and economic strategies continue to shape their futures as global powerhouses.

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